Andrew Brettin

New York, NY · brettin@cims.nyu.edu · Website · GitHub · LinkedIn

Climate Data Scientist with a PhD in Atmosphere-Ocean Science & Mathematics and 5+ years of experience in predictive modeling and uncertainty quantification. Expert in building machine learning algorithms for assessing weather and climate risk. Proven ability to manage TB-scale machine learning pipelines and communicate insights to stakeholders.

Relevant Experience

Courant Institute of Mathematical Sciences, New York University — New York, NY
Postdoctoral Research Associate — *May 2025–September 2025*
Graduate Research Assistant — *September 2019–May 2025*
Topics: Machine learning for quantifying climate risks and sea level predictability.

Mathematics and Climate Research Network, University of Minnesota — Minneapolis, MN
Research Assistant — *September 2017–May 2019*
Topics: Mathematical modeling of biodiversity in ecological systems.

Center for Discrete Mathematics and Computer Science, Rutgers University — New Brunswick, NJ
NSF Summer Research Intern — *May 2018–July 2018*
Topics: Stochastic models of cancer growth. Reaction-diffusion equations, Monte Carlo simulation.

Education

Ph.D., Courant Institute, NYU. Atmosphere-Ocean Science & Mathematics. Advisor: Laure Zanna — *May 2025*
B.Sc., University of Minnesota. Mathematics, *summa cum laude with high distinction (GPA 3.92)* — *May 2019*

Awards and distinctions: VoLo Fellowship (2020) • Henry M. MacCracken Fellowship (2019) • Hans H. Dalaker Scholarship (2018) • NSF REU Fellow (2018 & 2017) • Gold Scholar Award (2015)

Research Projects

Estimating temperature and precipitation extremes with quantile neural networks — *2025*

- Designed and benchmarked a novel machine learning framework for uncertainty quantification, outperforming industry-standard baselines across 1,500+ weather stations
- Engineered multi-source features by integrating gridded climate data (Google Cloud ERA5) with in-situ observations (NOAA GHCNh data), strategically combining reanalysis products with local measurements to improve predictive performance
- Validated model robustness using probabilistic evaluation metrics (CRPS, calibration), minimizing forecast bias and improving confidence interval reliability

Outcomes: [preprint] [code]

Learning efficient forecasts for regional sea surface height dynamics — *2024–2025*

- Customized a deep learning framework (convolutional autoencoders) with nonlinear dynamical systems theory (Koopman operators) to transform high-dimensional, complex spatiotemporal data into a tractable, linear latent representation
- Implemented Distributed Data Parallelism (DDP) in Pytorch to efficiently scale training over 200+ GB of spatiotemporal data on multiple GPUs
- Demonstrated superior predictive performance over stochastic Linear Inverse Models (LIM), improving forecasting skill by 5–10% using latent representations of similar complexity

Outcomes: [Article, *Geophysical Research Letters*] [code] [presentation]

Identifying sources of sea level predictability with mean-variance networks — *2023–2024*

- Architected scalable ETL pipelines to process TB-scale spatiotemporal climate datasets, leveraging Dask and Xarray for parallel processing on high-performance computing (HPC) systems
- Automated end-to-end training, optimization, and evaluation of an ensemble of 6,500+ neural networks

- Identified physical drivers of sea level predictability using Explainable AI (XAI) techniques

Outcomes: [Article, *Artificial Intelligence for the Earth Systems*] [code] [conference presentation]

Exploring nonstationarity of coastal sea level distributions *2021–2022*

- Developed a non-stationary statistical framework for quantifying shifting probability distributions in tide gauge observations, enabling more robust risk assessment of extreme events under shifting climate states

Outcomes: [Article, *Environmental Data Science*] [code] [presentation]

Technical Skills

Languages	Python (numpy, pandas, scipy, scikit-learn, matplotlib) • Bash • Julia • R • SQL • C++
MLOps	Linux • VCS (git/GitHub) • Virtual environments (conda) • Containerization (Docker/Singularity) • Pytorch/TensorFlow • Experiment tracking (Lightning AI/Weights & Biases) • Unit testing (pytest)
Computation	HPC clusters (SLURM/PBS) • Cloud platforms (GCP) • Distributed/parallel computing • GPU acceleration (CUDA/DDP)
Geospatial	Raster data strategies (xarray, Dask, ESMF regridding) • Temporal data strategies (cftime, flox) • File formats (NetCDF/HDF5/zarr/GRIB) • Visualization (cartopy/HoloViews)
Stats/Math	Uncertainty quantification • Extreme value analysis • Quantile regression • Bayesian inference • PCA • Unsupervised learning • Time-series forecasting • Numerical methods
ML	CNNs • Autoencoders • Random forests • Gaussian processes • U-nets • Diffusion models

Publications

1. **Brettin, A.** & Zanna, L. Estimation of temperature and precipitation uncertainties using quantile neural networks (preprint). 2026. arXiv: 2601.17243 `[physics.ao-ph]`.
2. **Brettin, A.** *Data-Driven Approaches for Predictability and Uncertainty Quantification of Sea Level Variability.* 3223742845. PhD thesis (New York University, 2025).
3. **Brettin, A.**, Zanna, L. & Barnes, E. Learning Propagators for Sea Surface Height Forecasts Using Koopman Autoencoders. *Geophys. Res. Lett.* **52,** e2024GL112835 (2025).
4. **Brettin, A.**, Zanna, L. & Barnes, E. Uncertainty-Permitting Machine Learning Reveals Sources of Dynamic Sea Level Predictability across Daily to Seasonal Time Scales. *Artif. Intell. Earth Syst.* **4,** 250014 (2025).
5. Falasca, F., **Brettin, A.**, Zanna, L., Griffies, S. M., Yin, J. & Zhao, M. Exploring the nonstationarity of coastal sea level probability distributions. *Env. Data Sci.* **2,** e16 (2023).
6. Meyer, K., Broda, J., **Brettin, A.**, Sánchez Muñiz, M., Gorman, S., Isbell, F., Hobbie, S. E., Zeeman, M. L. & McGehee, R. Nitrogen-induced hysteresis in grassland biodiversity: a theoretical test of litter-mediated mechanisms. *Am. Nat.* **201,** E153–E167 (2023).
7. **Brettin, A.**, Rossi-Goldthorpe, R., Weishaar, K. & Erovenko, I. V. Ebola could be eradicated through voluntary vaccination. *Royal Society Open Science* **5.** 10.1098/rsos.171591, 171591 (2018).

Additional Activities

- **NASA/JPL Summer School on Satellite Observations and Climate Models**, *Caltech, Pasadena, CA* *2023*
- **LEAP Momentum Bootcamp on Climate Data Science**, *Columbia University, New York, NY* *2022*
- **Volunteer tutor, math grades 5–8**, *Common Denominator, New York, NY* *2021–2022*
- **Vice President, Courant Student Organization**, *New York University, New York, NY* *2021–2022*
- **Project mentor, Undergraduate Research Program in Data Science** *2021*
  NYU Center for Data Science (collaboration with the National Society of Black Physicists), New York, NY
- **Reviewer**, *Geophysical Research Letters* (2025–) and *Journal of Geophysical Research: Machine Learning and Computation* (2025–)